

CHAMBERS'S JOURNAL

OF

POPULAR

LITERATURE, SCIENCE, AND ART

Fifth Series

ESTABLISHED BY WILLIAM AND ROBERT CHAMBERS, 1832

CONDUCTED BY R. CHAMBERS (SECUNDUS)

No. 35.—VOL. I.

SATURDAY, AUGUST 30, 1884.

PRICE 1½d.

A RIVER HOLIDAY.

WHAT the yacht-races at Cowes and a score of other places are to that section of the upper ten-thousand who delight in everything that pertains to the sea, and to whom the smell of salt water is as the breath of life—what Henley regatta is to those who find their exercise or pastime among the sunny reaches of the Upper Thames—such is the annual sailing-barge match from Erith to the Nore and back, to the vast river-side population below-bridge who have more or less to do, or are in some way connected, with the dock, canal, or up-river shipping traffic of the port of London. To these worthy people, as well as to some thousands of others from all parts of the metropolis, many of whom, in all probability, rarely adventure so far on the Thames at any other time, it is the race *par excellence* of the year; and it has much about it to render this widespread popularity deserved.

It is a bright midsummer morning, and the clock is on the stroke of nine when we find ourselves on Blackwall pier, with its vast shut-up hotel staring blankly across the river, once on a time famed far and wide for its capital fish-dinners; but now, alas, given over to desolation and decay. Even as far away as Dalston Junction, at which place we have to change trains, unmistakable signs of holiday-making are apparent; and at each station as we come along we pick up a numerous contingent, all of whom, to judge from appearances, like John Gilpin's wife, are evidently on pleasure bent.

We find the pier gay with summer costumes and smiling faces; friend greets friend after the hearty, robust English fashion which has not yet died out 'east of Temple Bar'; the river gleams with a thousand silver ripples in the morning sun; the heat is tempered by an exhilarating breeze; everybody prophesies that we shall have a glorious day. The majority of those on the pier are waiting the arrival of the excursion steamers from London Bridge. We, more fortunate than many, are the recipients of

an invitation to a private party which numbers, all told, some five-and-thirty souls. Presently, from among a cloud of others we single out the particular bit of bunting we have been told to look for; and there, at her temporary moorings at the upper end of the pier, we find the smart little *Cygnets*, our home that is to be for the next dozen hours. Old acquaintances welcome us with a cordial grip as we step on board, and new acquaintanceships are made, which in their turn will, we hope, grow ripper by-and-by. And now we have time to look about us.

The *Cygnets* is gay with bunting from stem to stern. Aft, a large awning is stretched, which will serve as a protection from either sun or rain, as the case may be. Camp-stools in abundance are provided, so that we can shift our quarters as we may list; and some neighbouring hotel has supplied us with several oblong mahogany tables, for which an excellent use will be found later on. Hampers crammed with good things solid and liquid are being brought on board one after another; and some one below deck is taking Time by the forelock already, in the way of putting a preliminary edge on the carving-knife. We are evidently going to have what our American cousins call 'a high old time' of it.

In confidence we may whisper that our little *Cygnets* is neither more nor less than a Channel steam-tug—one of that numerous fleet which scour the English Channel from the Lizard to the Languard, from the Mouse Light to Dunkirk, on the lookout for homeward-bound ships which, anxious to save a tide or two, and reach their moorings in dock as quickly as may be, are willing to pay for the help that will enable them to do so. A rough life, my masters, and not without its dangers when the stormy winds do blow. Often in wild mid-winter weather, or when the equinoctials seem as if they were tearing heaven and earth asunder, these little craft will remain out for days and nights together, afraid to risk making for any harbour, and preferring to keep in the open

CHAMBERS'S JOURNAL

OF
POPULAR

LITERATURE, SCIENCE, AND ART

Fifth Series

ESTABLISHED BY WILLIAM AND ROBERT CHAMBERS, 1832

CONDUCTED BY R. CHAMBERS (SECUNDUS)

No. 35.—VOL. I.

SATURDAY, AUGUST 30, 1884.

PRICE 1½d.

A RIVER HOLIDAY.

WHAT the yacht-races at Cowes and a score of other places are to that section of the upper ten-thousand who delight in everything that pertains to the sea, and to whom the smell of salt water is as the breath of life—what Henley regatta is to those who find their exercise or pastime among the sunny reaches of the Upper Thames—such is the annual sailing-barge match from Erith to the Nore and back, to the vast river-side population below-bridge who have more or less to do, or are in some way connected, with the dock, canal, or up-river shipping traffic of the port of London. To these worthy people, as well as to some thousands of others from all parts of the metropolis, many of whom, in all probability, rarely adventure so far on the Thames at any other time, it is the race *par excellence* of the year; and it has much about it to render this widespread popularity deserved.

It is a bright midsummer morning, and the clock is on the stroke of nine when we find ourselves on Blackwall pier, with its vast shut-up hotel staring blankly across the river, once on a time famed far and wide for its capital fish-dinners; but now, alas, given over to desolation and decay. Even as far away as Dalston Junction, at which place we have to change trains, unmistakable signs of holiday-making are apparent; and at each station as we come along we pick up a numerous contingent, all of whom, to judge from appearances, like John Gilpin's wife, are evidently on pleasure bent.

We find the pier gay with summer costumes and smiling faces; friend greets friend after the hearty, robust English fashion which has not yet died out 'east of Temple Bar'; the river gleams with a thousand silver ripples in the morning sun; the heat is tempered by an exhilarating breeze; everybody prophesies that we shall have a glorious day. The majority of those on the pier are waiting the arrival of the excursion steamers from London Bridge. We, more fortunate than many, are the recipients of

an invitation to a private party which numbers, all told, some five-and-thirty souls. Presently, from among a cloud of others we single out the particular bit of bunting we have been told to look for; and there, at her temporary moorings at the upper end of the pier, we find the smart little *Cygnets*, our home that is to be for the next dozen hours. Old acquaintances welcome us with a cordial grip as we step on board, and new acquaintanceships are made, which in their turn will, we hope, grow ripper by-and-by. And now we have time to look about us.

The *Cygnets* is gay with bunting from stem to stern. Aft, a large awning is stretched, which will serve as a protection from either sun or rain, as the case may be. Camp-stools in abundance are provided, so that we can shift our quarters as we may list; and some neighbouring hotel has supplied us with several oblong mahogany tables, for which an excellent use will be found later on. Hampers crammed with good things solid and liquid are being brought on board one after another; and some one below deck is taking Time by the forelock already, in the way of putting a preliminary edge on the carving-knife. We are evidently going to have what our American cousins call 'a high old time' of it.

In confidence we may whisper that our little *Cygnets* is neither more nor less than a Channel steam-tug—one of that numerous fleet which scour the English Channel from the Lizard to the Languard, from the Mouse Light to Dunkirk, on the lookout for homeward-bound ships which, anxious to save a tide or two, and reach their moorings in dock as quickly as may be, are willing to pay for the help that will enable them to do so. A rough life, my masters, and not without its dangers when the stormy winds do blow. Often in wild mid-winter weather, or when the equinoctials seem as if they were tearing heaven and earth asunder, these little craft will remain out for days and nights together, afraid to risk making for any harbour, and preferring to keep in the open

**The Project Gutenberg eBook of Chambers's Journal of
Popular Literature, Science, and Art, Fifth Series, No. 35, Vol.
I, August 30, 1884**

This eBook is for the use of anyone anywhere in the United States and most other parts of the world at no cost and with almost no restrictions whatsoever. You may copy it, give it away or re-use it under the terms of the Project Gutenberg License included with this eBook or online at www.gutenberg.org. If you are not located in the United States, you will have to check the laws of the country where you are located before using this eBook.

Title: Chambers's Journal of Popular Literature, Science, and Art, Fifth Series, No. 35, Vol. I, August 30, 1884

Author: Various

Release date: August 25, 2021 [eBook #66141]
Most recently updated: October 18, 2024

Language: English

Other information and formats: www.gutenberg.org/ebooks/66141

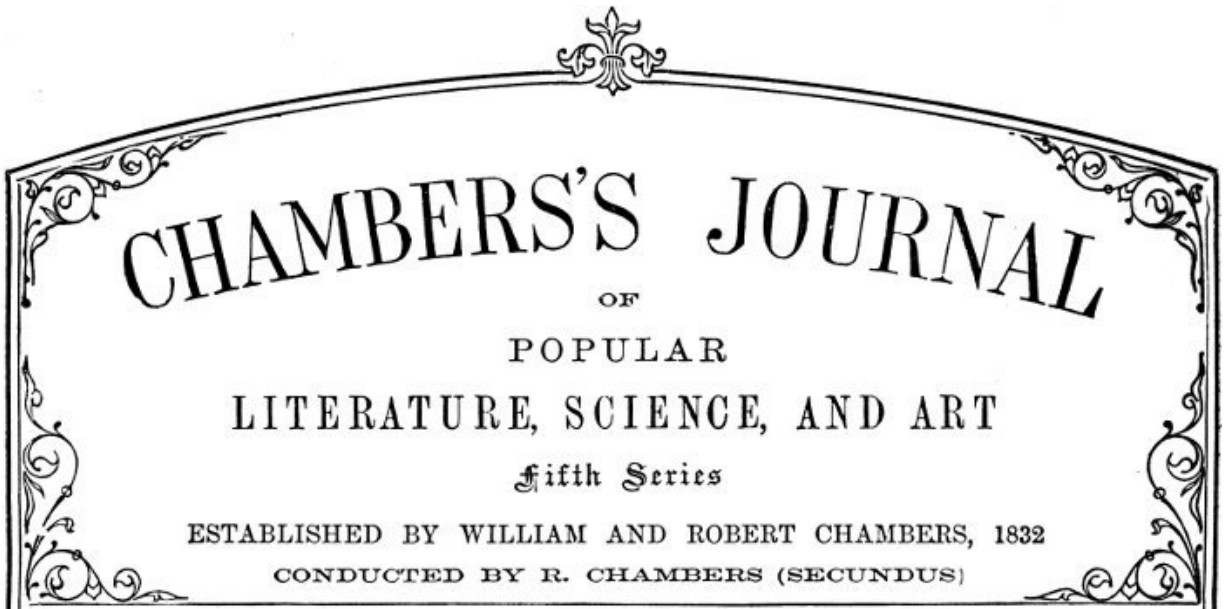
Credits: Susan Skinner, Eric Hutton and the Online Distributed Proofreading Team at <https://www.pgdp.net> (This file was produced from images generously made available by The Internet Archive)

*** START OF THE PROJECT GUTENBERG EBOOK CHAMBERS'S
JOURNAL OF POPULAR LITERATURE, SCIENCE, AND ART, FIFTH
SERIES, NO. 35, VOL. I, AUGUST 30, 1884 ***

**CHAMBERS'S JOURNAL
OF
POPULAR
LITERATURE, SCIENCE, AND
ART.**

CONTENTS

A RIVER HOLIDAY.
BY MEAD AND STREAM.
HOME-NURSING.
VERMUDYN'S FATE.
QUEEN MARGARET COLLEGE.
THE MONTH: SCIENCE AND ARTS.
OCCASIONAL NOTES.
EN PASSANT.



No. 35.—VOL. I.

SATURDAY, AUGUST 30, 1884.

PRICE 1½d.

A RIVER HOLIDAY.

WHAT the yacht-races at Cowes and a score of other places are to that section of the upper ten-thousand who delight in everything that pertains to the sea, and to whom the smell of salt water is as the breath of life—what Henley regatta is to those who find their exercise or pastime among the sunny reaches of the Upper Thames—such is the annual sailing-barge match from Erith to the Nore and back, to the vast river-side population below-bridge who have more or less to do, or are in some way connected, with the dock, canal, or up-river shipping traffic of the port of London. To these worthy people, as well as to some thousands of others from all parts of the metropolis, many of whom, in all probability, rarely adventure so far on the Thames at any other time, it is the race *par excellence* of the year; and it has much about it to render this widespread popularity deserved.

It is a bright midsummer morning, and the clock is on the stroke of nine when we find ourselves on Blackwall pier, with its vast shut-up hotel staring blankly across the river, once on a time famed far and wide for its capital fish-dinners; but now, alas, given over to desolation and decay. Even as far away as Dalston Junction, at which place we have to change trains, unmistakable signs of holiday-making are apparent; and at each station as we come along we pick up a numerous contingent, all of whom, to judge from appearances, like John Gilpin's wife, are evidently on pleasure bent.

We find the pier gay with summer costumes and smiling faces; friend greets friend after the hearty, robust English fashion which has not yet died out 'east of Temple Bar;' the river gleams with a thousand silver ripples in the morning sun; the heat is tempered by an exhilarating breeze; everybody prophesies that we shall have a glorious day. The majority of those on the pier are waiting the arrival of the excursion steamers from London Bridge. We, more fortunate than many, are the recipients of an invitation to a private party which numbers, all told, some five-and-thirty souls. Presently, from among a cloud of others we single out the particular bit of bunting we have been told to look for; and there, at her temporary moorings at the upper end of the pier, we find the smart little *Cygnet*, our home that is to be

for the next dozen hours. Old acquaintances welcome us with a cordial grip as we step on board, and new acquaintanceships are made, which in their turn will, we hope, grow riper by-and-by. And now we have time to look about us.

The *Cygnet* is gay with bunting from stem to stern. Aft, a large awning is stretched, which will serve as a protection from either sun or rain, as the case may be. Camp-stools in abundance are provided, so that we can shift our quarters as we may list; and some neighbouring hotel has supplied us with several oblong mahogany tables, for which an excellent use will be found later on. Hampers crammed with good things solid and liquid are being brought on board one after another; and some one below deck is taking Time by the forelock already, in the way of putting a preliminary edge on the carving-knife. We are evidently going to have what our American cousins call 'a high old time' of it.

In confidence we may whisper that our little *Cygnet* is neither more nor less than a Channel steam-tug—one of that numerous fleet which scour the English Channel from the Lizard to the Languard, from the Mouse Light to Dunkirk, on the lookout for homeward-bound ships which, anxious to save a tide or two, and reach their moorings in dock as quickly as may be, are willing to pay for the help that will enable them to do so. A rough life, my masters, and not without its dangers when the stormy winds do blow. Often in wild midwinter weather, or when the equinoctials seem as if they were tearing heaven and earth asunder, these little craft will remain out for days and nights together, afraid to risk making for any harbour, and preferring to keep in the open while waiting for the gale to blow itself out. Only a few winters ago, as the *Napoleon* steam-tug was towing a huge liner up the Channel in the teeth of a tremendous storm of wind and sleet, her hawser parted, and when, some two minutes later, the crew of the big ship had time to look for the tug, she was nowhere to be seen. At the moment her hawser broke she had been struck by a heavy sea, and had plunged down head foremost, she and all her crew. As a rule, however, these sturdy little craft, which are built as strongly as iron and timber can make them, will safely ride out a gale such as might well cause many a big merchantman to quake with fear.

But to-day all thoughts of storm and disaster are far from us as we sway gently at our moorings in the morning sunshine and watch the ever-

changing panorama before us. The twin domes of Greenwich Hospital show white and ghostlike through the faint haze which veils everything in the distance. Presently round a point of land where the river curves sharply away to the left comes gliding in stately fashion the big saloon steamer *Alexandra*, followed by the *Albert Edward* and several smaller boats, all with numerous flags flying fore and aft, and all, or nearly all, with bands of music, military or otherwise, on board. A few minutes later, the Committee Boat, the old and well-known *Eagle*, puts in an appearance. Suspended by knots of blue ribbon from the captain's bridge are the silver cups which will be competed for a little later on. More passengers crowd aboard the big steamers; one of the bands plays lustily, an irrepressible drummer being well to the fore; flags flutter in the breeze; our moorings are cast off; the *Cygnets* gives one last screech of triumph, or, it may be, of farewell to those left behind, and at length we are fairly off on our way to Erith. In front of us, behind us, and on either side of us are steamers and tugs of all kinds and sizes; but the river is wide; there is room enough for all, and we steam along in pleasant company. Now is the time to make sure of a little luncheon, so that we may not miss the start of the race later on.

Erith is reached a little before eleven; and here we find the competing barges arranged in order, waiting for the signal, while the pleasant little town itself is *en fête*, and thousands of eyes are looking on from the shore. We voyagers who have come to watch the race keep well in the background, so as not to impede the start; the Committee Boat takes up its position; a gun is fired; and before you know what has happened, anchors are tripped, sails are loosened to the breeze, and the barges, topsail and spritsail, spring forward on their course like a flight of dark-hued seabirds newly set free.

The topsail barges—so the official programme informs us—are not to exceed fifty-five tons register. The first prize is a silver cup of the value of twenty pounds, and ten guineas for the crew; the second prize is a silver cup of the value of fifteen pounds, and five guineas for the crew; then follow other prizes of lesser value. The spritsail barges are not to exceed fifty tons register. The prizes follow in the same ratio as those for the topsails, but are not quite equal to them in value. In addition to their money prizes, a champion flag is presented to each of the winners, which will flaunt proudly

in the breeze on their voyages up and down the river for many a day to come.

There are fourteen competitors in the race this year, namely, eight topsails and six sprits. The topsails, merely because they are topsails and spread more canvas to the breeze than the others, gradually forge ahead; but that is only what everybody knows will happen. Having seen them fairly under way, we steam gently along, pass through the midst of the little fleet, and then get well ahead of them, but not so far as to be altogether quit of their company, except when some sharp bend in the river hides them for a little while from view. Now is the time to get up a friendly sweepstake on board, a task which two of the company undertake, and carry out satisfactorily. Some enthusiasts have discovered a pack of cards, and are already deeply immersed in the intellectual game of Nap. By-and-by, the old historical fort of Tilbury is reached and passed; and before long, Gravesend comes into view with its famed hotels and its Gardens, at which we have been so frequently reminded we may 'spend a happy day.' Here we come to a stand for a little while, in order that we may watch the procession pass, as do many of the other gaily-decked tugs, together with some of the big steamers. It is a pretty sight to watch the brown and chocolate coloured sails come stealing round the reaches of the river, and to see how cleverly the little craft are handled as they tack here and there to catch an extra capful of the capricious westerly breeze, or to steal for a few moments the wind out of some rival's sails. To-day, in honour of the occasion, the crews are rigged out in new blue jerseys and knitted scarlet caps; while the boats themselves are as spick-and-span as paint and gilding can make them. Each barge carries at its fore its official number on the programme; and as they glide one by one into view, innumerable are the glasses levelled at them in the effort to make out either their name or number. But position in the race at this point is held to be of small account by those who are supposed to be knowing in such matters: Tattenham Corner—otherwise the lightship at the Nore—is still a long way ahead.

Again we steam along in the wake of the barges, again pass through the midst of them, and again leave them astern. For a while we have left behind us the excitement of the race. There is a pleasant sound of the drawing of corks. It is the time for a cigar, a chat, and a bottle of Bass. As we go gently down, we pass several heavily-laden barges making their way up river,

some of which are pointed out to us as winners in matches of years gone by; but their racing-days are over for ever, and they have evidently settled down to the sober, steady work of middle age. They hail chiefly from the Medway district, we are told, and are laden with cement, lime, bricks, stone, hay or straw, some of them voyaging as far inland, by way of the Regent's Canal, as Camden Town and Paddington.

And so after a time Southend comes into view, with its terrace-crowned cliff looking far out across the river, and its mile-and-a-quarter-long pier, which seems as if it were stretching out a friendly hand to greet its neighbour, Sheerness, over the way. Half an hour longer brings us to the Nore.

The lightship at the Nore is the point round which each barge has to make its way before starting on its return journey up river—the distance in all, so we are informed, being about seventy miles: not a bad day's work for a class of craft which many people are in the habit of decrying as the tortoises of the river. Occasionally it happens that there is not enough wind to enable them to complete the regulation course, in which case the Committee on board the *Eagle* have power to fix the point at which the return journey shall begin.

We have been taking matters easy for the last hour or so, and we find several steamers and tugs lying on and off round the lightship when we reach it. We follow their example, keeping up just enough steam to prevent us from drifting with the tide, and here we are presently joined by other steamers and pleasure-craft of various kinds. Among the rest comes the indispensable Committee Boat, which is moored alongside the light. Not long have we to wait before the cinnamon-coloured sails of the little fleet steal into sight one by one. Glasses come into requisition again, and all are agog to make out the number of the leading topsail. 'No. 3—*Frances*,' calls out some one keener-sighted than the rest. And so it proves to be. Gallantly she comes sweeping down, every man at his post, every eye on the alert. Suddenly the helm is put about; we see the crew hauling at the ropes like red-nightcapped demons, and then we hear the swish of the water as the *Frances*, answering to the call upon her, sweeps round the lightship in a short but graceful curve, and catching the breeze next moment on the opposite tack, is speeding away on her return journey, followed by a ringing cheer from a thousand throats.

The next to round the Nore light is the *Whimbrel*, and after her comes the *Bras-de-Fer*; while the leader of the spritsails, or ‘stumpies’ as they are familiarly called, is the *Bessie*. We do not wait till the whole of the laggards have rounded the light, but steam gently away till we come to a certain quiet, sunny reach, where we lie by while sundry hampers are opened and a large measure of justice is meted out to their welcome contents. After this pleasant interlude, onward again at full speed till we once more catch up the barges. Now does the excitement grow apace among such of us as have drawn fortunate numbers in the sweepstake, to ascertain which are the leading craft, for as their positions are by this time, such in all probability will they be at the finish.

Still we go pulsing along at a great pace, showing our heels to many a steamer as big again as ourselves, till at length we find ourselves once more at Erith. Here we secure a position close to the Committee Boat, and not far from the winning-post—a small buoy with a flag atop of it anchored out some distance in the river. Gradually more steamers and tugs take up positions no great distance away. On every side of us are music, dancing, feasting, and high-jinks generally; but not one angry word, not one coarse expression is anywhere to be heard. Nowhere could there be a better-tempered holiday crowd.

At length a buzz, a murmur, a general movement, and each one says to his neighbour, ‘Here comes the first topsail,’ while everybody seems to ask at once, ‘What’s her number?’ A gun is fired, a band plays *See the conquering Hero comes*, a great shout is set up, and we all know that the *Whimbrel* has won the first prize.

‘Never prophesy till you know,’ seems to be a maxim of wide application. Who would have thought that the saucy *Frances*, which headed all the others round the Nore, would only come in fifth at the finish? But so it was; while the *R. A. Gibbons*, which was fourth round the light, came in for the second prize. So among the spritsails—the first round the light came in second, and the second first.

We do not wait to see the prizes given away, for the evening is growing chilly, and many of us have a long way to go. We chase the dying sunset as we steam swiftly up stream, but fail to overtake it. Little by little its splendours soften, fade, and vanish. Some time between nine and ten, and

while there is still a dusky shimmer on the river, we find ourselves once more at Blackwall pier; and there we part, hoping to meet next year when time shall have again brought round the pleasant River Holiday.

BY MEAD AND STREAM.

CHAPTER XLIV.—AN APPLE OF DISCORD.

It did not occur to either of these young people that there was anything at all remarkable or irregular in the circumstance of a lady visiting the chambers of her betrothed alone. But as this was her first visit, Madge felt a little awkward, and would have been much more at ease if Wrentham had not been present.

That gentleman, however, as soon as he perceived who the visitor was, took up his glossy hat, made his salutations to Miss Heathcote, and informed Philip that he was obliged to hurry along to the office before it closed, but would probably return later.

When he had departed, Madge glanced with curiosity round the apartment, and her first comment was:

‘You ought to have curtains over that doorway, Philip’ (she alluded to the uncovered entrance to a small recess which was a storeroom); ‘and I must come in soon and dust the place thoroughly. I wonder you have not been choked. See here; it is positively disgraceful.’

She ran her finger over the ledge of a bookcase, making a line in the dust. And with half-timid but wholly curious interest, she continued to scrutinise the place, making mental notes of what she would have to do to insure his comfort.

He was astounded. She had been with Mr Shield. She must have been made acquainted with the terrible nature of his position; and yet she could placidly criticise the furniture of his room and interest herself in a question of dusting! He had often admired her cool firmness in moments of accident, illness, or difficulty; but he could find nothing to admire in this absolute indifference to the crisis in his affairs. In his bitterness he was unjust, and his reflections were to this effect: ‘How blessed are those who can be callously calm in the presence of suffering—of the suffering even of those they are supposed to love! How many pangs they must be spared; how easy

it must be for them to pass comfortably through the world, where every step we take leads us by some scene of misery. Ay, they are the happy ones who can pass with eyes closed, and therefore, nerves unshaken.'

But even whilst these uneasy thoughts were flashing through his brain, he felt ashamed of himself for allowing them to be suggested by Madge, whose calmness he knew was not due to want of feeling, but to a delicate shrinking from the display of it.

She appeared to become suddenly aware of his singular silence, and looked quickly towards him. His face was in shadow, and she could not see the ravages which anxiety and sleepless nights had made upon it; and he did not observe that under her apparent composure there was suppressed much agitation. The tender eyes looked at him wistfully, as if afraid that she had done something to offend him, and that he was about to chide her.

'Why do you not speak, Philip?'

'I was wondering if it can be possible that you have not heard how things stand with me. I was at Willowmere this forenoon, seeking you, and was told that you had gone to see Mr Shield, intending also to call on me. Has he said nothing to you about the letter I sent to him last night? I was obliged to write, because he persists in refusing to listen to any explanations from me in person. Has he said nothing about it?'

Madge hesitated. She was in a most unpleasant position. She had hoped to be able to come gleefully to him with the good news that the reconciliation between his father and uncle had been effected, and she was disappointed. Her proofs of Mr Hadleigh's innocence of all complicity in Austin Shield's misfortunes had not been accepted in the way she had expected. As regarded Philip, she had been assured that he was safe so long as she kept her promise to Mr Beecham. So she could neither give him the good news she had been so confident of bringing to him, nor sympathise fully with his anticipations of absolute ruin. That was what rendered her manner peculiar, and in his present vision, ungracious.

'I have been told that you are harassed by the way things have been going, and that there have been mistakes somewhere. But I heard nothing about your letter.'

'And yet you have been with him and Mr Beecham all day!'

She did change colour at the mention of Beecham's name, the blood flushing her cheeks, and then as suddenly fading from them. His overwrought nerves rendered him sensitive to the slightest change of voice, look, or manner.

'Yes,' she replied at length steadily; 'I have been with them a long time to-day, and they spoke a great deal about you, for they are both your friends.'

'No doubt, no doubt. Beecham has no reason to be otherwise; and Mr Shield has acted as my friend until now, when he leaves me in this horrible suspense.'

'But it must be because he is considering what is best to be done for you.'

'Did he tell you that?'

'He did not say it exactly in those words; but I understood it from what he did say and from his whole manner in speaking of you.'

'I suppose I ought to find satisfaction in that.... But how was it you came to visit Mr Shield? You have not met him before.' (This abruptly.)

Her eyelids drooped, and her head was bowed a little.

'He wrote to me. I have met him before.'

'And you never told me! Where did you become acquainted with him?'

'At Willowmere.'

'Why, when was he there? Aunt Hussy does not know of it, or she would have told me. You did not, although you should have known how pleasant it would have been to me to find that he had seen you and liked you.'

That she had not previously told him of her acquaintance with Mr Shield, was a disagreeable sign of want of confidence; but his surprise was greater than his displeasure. He had never been able to obtain more than ten or fifteen minutes' audience of him; and yet here was Madge, without giving the slightest hint that she had ever seen him, accepted by him as a friend, and allowed to spend hours with him. If this was not deception on her part, it bore such a strong resemblance to conduct of that kind as to make him feel cold. A new pain entered his distracted mind. If she were capable of deceiving him in one way, how was he to trust her in other ways? She knew how he hated all mysteries and underhand work. She knew how he insisted

on the simple rule, that as it was so much more easy and comfortable to be plain and above-board in everything, than to adopt subterfuge, only fools chose the crooked course. Yet here he found that, for some unknown reason, she had been concealing most interesting facts from him.

To Madge the conversation was becoming more and more awkward and even distressing. She could feel the suspicions which were hovering around him, and she made an effort to dispel them by assuming a hopeful and, as far as possible, a cheerful tone.

‘Well, Philip, he asked me to hold my tongue because he wanted to give you a surprise; and I do not see any harm in it. Will you not let me have a little freedom of action, when I think I am doing what is to your advantage?’

‘There never can be any advantage gained for me by your hiding things from me.’

‘But you must not look upon it so seriously, Philip,’ she said with a mingling of earnestness and playfulness. ‘Come now; let us talk about what is of most importance to us both. Tell me how it is your affairs have come to such a crisis so soon, and how you mean to proceed.’

‘I shall do so; but first I must ask you how long Beecham has known Mr Shield?’

‘A long time,’ she replied, averting her eyes.

‘And has the secret he confided to you anything to do with me or my business?’

She would have liked to answer at once, and she was obliged to hesitate. She saw that he was vexed, and her natural impulse was to remove every source of vexation between them by telling him all she knew. The impulse was restrained on his account.

‘It has to do with you; but I wish you would not press me on the subject—at least not for a little while.’

‘So be it. I have always respected your wishes,’ he rejoined coldly, and there was even a distant note of bitterness in the tone. ‘I can now easily give you the information you require about myself. Should my uncle decline to assist me, I shall to-morrow resign everything I possess to my creditors, and

seek some employment by which I may be able in course of time to make up to them whatever deficit there may be in my accounts.'

'But Mr Shield will assist you—he will not allow you to give up everything!'

'As you will not permit me to know the grounds of your confidence in the continuance of his generosity, and as I have bitter reason to know that he would be justified in refusing to give further help to a fool who has in such a short time made away with the capital he placed at my disposal, I cannot share your expectations or hope.'

'I am sure he will carry you safely over this difficulty.'

'In any case, I am his debtor, and the necessity to repay him'——

'But he does not expect you to repay him,' she interrupted, watching him with rapidly increasing anxiety, and now observing how haggard he looked.

'I *will* repay him,' was the answer, emphasised by passion that was suppressed with difficulty. 'I know it will take a long time—maybe all my life. Knowing that, I am compelled to regard as inevitable and just the view which Mr Crawshay will take of our position. He will insist on the same arrangement which he insisted on when I intended to go abroad.'

Wonder was in her eyes, strange pain in her breast. She could scarcely remember the time when, except in the presence of strangers, Philip had spoken of Uncle Dick as Mr Crawshay. This simple change affected her more than his words or his manner, for he maintained a degree of the bitter calmness of despair. There must be some evil at hand greater than she could imagine, since it forced him to refer to his friend at Willowmere in that way.

'What arrangement are you speaking about, Philip?'

'I agreed to it then with a light heart; I agree to it now with a hopeless one. Then it was a jest—now, it is earnest. But it was wise, and it is wiser now. He required me to consider our engagement at an end, and to leave you free to choose'——

'Oh, Philip, Philip!'

The cry came in such piteous accents, that despite his frenzy he stopped. For a moment he was conscious of the cruelty he was perpetrating in

making such an announcement so abruptly. The golden visions of the future they had so often conjured up together flashed through his mind, and he was dazed with pain like her own.

For Madge, she had covered her face with trembling hands, as if in that way she could shut out the thoughts his words suggested. 'Free to choose some one else,' was what he had been going to say, she knew. Free! Could love be ever freed when once given? He might die before her; then she would live on his memory. He might go away from her and never return; what difference could that make? Men change; women change; but the being once realised in the idealism of love never changes to the lover. Else how could love survive, when the mortal form becomes plain and ugly, old and petulant?

Her thoughts did not run precisely in this form, but they were to the same purport. She could never care for any man but Philip; and to suggest the possibility of it would have been hard to bear if made by any one, but hardest of all when made by Philip. Then a little spring of mingled indignation and pride started, and the hands dropped from her face.

'And can you think that any one at Willowmere would turn from you at a time of trouble?'

'No, no; I do not mean that,' he answered, and his voice had become feeble, whilst his body swayed slightly, as if he were struggling with diverse emotions. 'But if it was fair that you should not be bound down to a man who was only going away for a year, it cannot be fair to bind you to one who may have to contend with poverty all his life.'

'Mr Shield—your father will see that it is not so.'

These names roused him, and his thoughts became collected again. He spoke almost calmly.

'My father has distributed his fortune amongst his other children. Mr Shield has given me a fortune which I, by my careless folly, have squandered or allowed myself to be cheated out of, as a fool in a betting-ring might have done. I must pay the penalty of my folly alone. Therefore I say, you are free.'

She took the lamp and held it up so that the full light fell on his face. There was a wildness in his eyes, but his lips were compressed, as if he had come

to an unalterable resolution.

‘Do you *wish* me to think myself free?’—the voice steady, although the lips trembled.

‘I wish it!’

A pause; and presently through the silence came the low sad words:

‘Then we must say Good-bye.’

‘Good-bye’ was the husky response, and that was all.

(To be continued.)

HOME-NURSING.

BY A LADY.

SECOND ARTICLE.

BEFORE commencing our subject proper, the sick-room, it may be well to consider two points very frequently neglected in home-nursing. First, as to a nurse's dress. Unless the case be infectious, nothing is better than some soft woollen material that will not rustle or creak, after the fashion of silk or print, but that will bear washing should the necessity arise. If the patient's taste is known and can be consulted, all the better; but if a favourite dress is too valuable to be devoted to sick-room wear and tear, a ribbon bow of some soft bright colour, and spotless collar and cuffs, will help to give that air of quiet cheerfulness which is soothing to senses so often rendered painfully acute by illness. Should there be more than one patient to attend to, or should the one be quite helpless, there will be a considerable amount of injury to clothing by rubbing against the beds, &c., which probably accounts for the style of dress affected by the professional nurse, which consists usually of a costume of either black flannel or stiff print. The former is so unsuitable, that it may be regarded as amongst the last relics of barbarism; and the latter, though economical and clean-looking, has the great drawback of creaking to an unlimited extent, and, moreover, would give the home-nurse an unnatural appearance—a thing to be studiously avoided.

As regards economy, a good substitute for a costume bristling with starch will be found in a large apron with a full bib, and loose sleeves to draw up and tie over the elbows. Even these should be made of a pretty and soft material; for, in our experience, colour and cheerful surroundings seldom fail to exert a beneficial influence. As an instance of the decided effect of colour, take the case of a baby, who at six months had taken no notice whatever of his surroundings; his parents were beginning to fear the possibility of blindness, when a friend coming in one day wearing a bright necktie, the sober little face relaxed, and a smile brought expression to the

hitherto vacant features. The fact was little Hugh had never seen anything but black on his nurses, and the sight of a bit of bright colour woke up new ideas of pleasure. I have said that illness often brings back much of the sensitiveness of childhood, and for this reason, in dealing with the sick, even small details are worthy of careful consideration. As to what a nurse should wear on her feet, there are few people who would not be horrified at the idea of creaky shoes; but I am by no means sure that the popular notion of list slippers for sick-room use is not a worse evil. Any one who has experienced the sensation of being wakened by a sudden presence at his bedside, can see how injurious must be the same experience to the invalid, who is in a state far more susceptible to shock, and who, once frightened, will not easily lose the dread of a repetition. So, on these grounds, wear only ordinary house-slippers without heels; and in walking across a patient's room, be careful to tread quietly, but at the same time in a firm, even way, and never on tiptoe, nor in that elaborately slow, hesitating manner which keeps an invalid on tenter-hooks of anxious watching.

Our second point—the care of a nurse's own health—is one on which it is impossible to strike too serious a note of warning, for important as it is, there are very few who give it practical consideration. Yet, over-zeal is sure to defeat itself, and nature, the sternest balancer of accounts, only allows a certain amount of work to be done, and rigidly exacts the penalty from those who forget or ignore her wise limitations.

All institutions sending out nurses have fixed rules as to a certain number of hours for sleep and exercise, without which, experience teaches, no one can safely carry on the laborious duties of a sick-room; yet the inexperienced imagine they can do what the trained nurse wisely refuses to undertake, and make attempts at such work as nursing both by night and day. Such attempts generally retard the patient's recovery, and always cause more or less injury to the nurse whose zeal has been without knowledge. In all cases where the patient is ill enough to need night-watching, two nurses are absolutely needful; but one may with advantage take the lead, and never leave the patient without arranging that he shall be properly cared for in her absence. The strongest, physically, had better be chief; and it will be well if she can undertake the whole of the night-work.

It is this question of night-work that is the *bête noire* of inexperience; but properly managed, and given an average amount of health, there is no

reason why there should be any great fatigue, even with prolonged night-watching. The one essential thing is, to understand and remember that there *must* be a good allowance of sleep, and at least two hours devoted to brisk, open-air exercise. It is one of the rarest things to find the latter point remembered in amateur nursing, and I have known cases where the whole female portion of a family has remained indoors for weeks, simply for want of understanding the vital importance of fresh air and exercise to counterbalance the unaccustomed strain of nursing. No wonder that in such cases, depressed spirits and shaken nerves become associated with night-nursing, when, as a matter of fact, it is only ill-regulated zeal that is to blame.

Still, at first, night-nursing does seem formidable, especially when, as often happens, it is made to follow upon an anxious day. The only wise method of beginning is to lie down in the afternoon, after a warm bath if possible, and try to read yourself to sleep. If you fail, the rest itself will be some preparation; and if you succeed, you will be surprised to find how easy your work will be. Take a good meal, and wash your hands before going into the sick-room; but do not commence work before eleven o'clock at the earliest. Beginning night-work too early is a mistake, especially where there is a natural tendency to fall asleep under the influence of warmth and quiet; but by making it as late as eleven or half-past, you will have a much better chance of keeping awake without a struggle. Ordinarily, too, a nurse not going on duty early will be able to take the lead in washing the patient in the morning and in making his room tidy. When this is done, she should give directions for the day, and, if possible, not enter the sick-room again till it is her turn to mount guard. The only drawback to this plan is that there may be difficulty in arranging to meet the doctor; but a little management will generally smooth the way, especially if helpers are reliable.

On leaving the sick-room, the night-nurse should at once go for a brisk walk, if possible with a pleasant companion, and the walk ought to occupy a couple of hours; but if exercise has not been a habit of life, it will be well to begin with less and gradually increase. It must be remembered that a dawdling lounge is useless, and that the walk must be brisk to be of any real service. On returning, the nurse should at once go to bed and have her sleep out. But if she feel particularly wide awake, a warm bath will supplement the effects of exercise. On waking, she should take a cold or tepid bath

according to habit. A nurse should be careful to change her under-linen as often as convenient.

One other thing must be borne in mind in regard to night-work, and that is, the necessity for taking food during the hours of watching. A nurse who takes proper time for sleep, misses at least one meal in the day, whilst needing more than the ordinary allowance of food; so that it is her duty to take nourishment during the night. A meal between two and three will help her through the hardest part of the twenty-four hours; and as soon as she feels hungry or weary, a glass of milk with an egg in it, a cup of cocoa, or some light soup, will give the needed support, and will also make a great difference to the ease of keeping awake and on the alert.

If these rules are carefully followed out, we venture to say there will be very little cause to dread even the most trying part of nursing—night-work.

And now as to the sick-room itself. If a choice is possible, let the room selected be of good size, cheerful, and quiet. It needs to be fairly large, because air is consumed by nurse as well as patient; for this reason, a dressing-room adjoining is of great service. Except in acute and dangerous illness, it is better if the nurse can sleep away from the patient, always provided there is ready means of communication. Helpless patients, as a rule, have a natural dread of being left alone; but few will object to a nurse's going to bed in an adjoining room, as long as they have the means of calling her at a moment's notice. If she be a light sleeper, a piece of tape tied to her wrist, the free end being left within easy reach of the patient, will be enough; or instead of tying the tape to her wrist, she may fasten a small bell, letting it rest over the head of her bed. Where the patient is very weak, an excellent contrivance is a piece of india-rubber tubing with a whistle at one end, and a compressible air-ball at the other. The latter should be placed on the patient's pillow, and by the slightest possible effort, he will be able to make the whistle sound. Of course, a nurse who adopts such methods must have dressing-gown and slippers at hand, that she may obey the summons instantly, for nothing is more likely to irritate a patient than being kept waiting at night.

The sick-room should, if possible, face south or south-west, so as to get the benefit of the sun. Should the light be too strong, it is easily regulated by drawing down the blinds, or by hanging up a piece of some dark material;

and in convalescence, the cheerful light of the sun plays an important part. In a sunny room, however, it is necessary to exclude the early morning light. The rising sun begins—in summer—to shine just at a time when, if the patient sleeps at all, he will be most likely to doze off; and it need hardly be said that to allow him to be awakened then is to deprive him of one of his best chances of improving.

But whilst cheerfulness is an essential of a sick-room, it is hardly less important that it should be free from liability to sudden noises. It should, therefore, never face a thoroughfare; nor, in a large family, be so situated as to necessitate much in the way of footsteps overhead. In cases where there is a nursery, it is well to take that for the invalid, at whatever risk of injury to other rooms; for nothing can be more distressing to a patient's nerves than the constant pitter-patter of small feet, added to the tumbles and screams inseparable from nursery-life. At the same time, a room at the top of the house has the serious disadvantage of causing much extra up-and-down-stair work, so that in small, grown-up families, it is well to choose a room as low down as possible. In houses where there are bedrooms behind the sitting-rooms, it is convenient to take one of such, especially where there is the comfort of a slab outside, of the use of which we shall have more to say later on.

A sick-room should not have French-windows, those opening at the top and bottom being much better for ventilation; and if possible, there should be either venetian or sun blinds, for the easier regulation of light. Before beginning to nurse a case, it should be ascertained that bolts and sashes of windows, cords and pulleys of blinds, hinges of doors, and ventilators, are all working easily and quietly. It would seem hardly necessary to add that a fireplace with a good grate, and a chimney that does not smoke, are also essential.

Before removing a patient to the room that has been selected, it should be well cleaned, the doors and windows left open, and a fire lighted. By the time the latter has burned up brightly, the air of the room will be perfectly fresh, and one of the nurse's first considerations will be how to keep it so. Her aim should be so to arrange ventilation that at no time should an incomer perceive any closeness or smell; in other words, the air of the sick-room ought to be as pure and fresh as the outside air; but in our climate this

is not always easy, and will never be accomplished without constant thought and attention.

To understand how important is the question of ventilation, it is well to consider what it is that causes air to become impure, and consequently unwholesome. The air we breathe consists, roughly speaking, of two gases, oxygen and nitrogen. The former is absolutely necessary for the maintenance of animal life; it is drawn into the lungs, to be mixed with the blood, and used in various operations of the body; consequently, the pure gas of oxygen becomes used up, and the air we breathe out has changed its character, and is charged with the poisonous gas, carbonic acid. If the same air continues to be breathed over and over, carbonic acid increases its proportions with each inspiration, and fatal results follow. From this it will be seen that even in health there is positive necessity for providing a supply of fresh air, as well as for the removal of that which has become vitiated. But in sickness, the need is even greater, as the air given off from the lungs of the invalid will contain a larger proportion of poisonous matter. Impure air has always a tendency to ascend, and the secret of successful ventilation consists in getting rid of the warm, vitiated air floating at the top of a room. Once get this out, and nature, abhorring a vacuum, will pour in a fresh supply of pure air. You may sometimes be able to do this by opening the window wide for a few minutes, your patient being meanwhile covered up with an extra blanket, and a light handkerchief over his face. But in cold weather, this would lower the temperature of the room too much, and in any case it is hardly a method for the unprofessional nurse, unless with the doctor's special permission. And even if this is allowed, it will not be enough, as the air consumed by you and your patient requires constant as well as thorough changing.

A fire is an excellent ventilator, as by it warm air is constantly being drawn up the chimney, whilst its place is taken by fresh; but a fire alone will not be sufficient, though it will enable you, often, to keep the window open an inch or two. This will in ordinary cases be quite sufficient; but it often happens that a patient unused to fresh air complains of the draught of an open window, and asks to have the door open instead. Never, if possible, yield to this. It is one of the commonest mistakes in home-nursing. As I said before, impure air ascends; and so, if your room be above the ground-floor, the heated, vitiated air from all the rooms below will come pouring through the

open door of the sick-room. Yet, I have known cases of long-standing illness where there has been no attempt at ventilation other than through the door, and where the window has not been opened for months. In such cases, it not seldom happens that nurses complain of feeling heavy on waking—they and the patient have been using up the same air all night—and yet obstinately refuse to put the window down or use a ventilator, or even to see that the staircase window near the sick-room is kept open. The last-named expedient is the only one by which door-ventilation can be of any use; but it is far better to adopt one of the following plans, nearly always available and safe, even for the most delicate. The first is a very simple contrivance, which deserves a place in every bedroom where the window is not kept open at night. Get a piece of wood the exact width of the window and have it nailed to the lower sash; you will then have a space between the two sashes, through which cold fresh air will enter; the current will drive it up towards the top of the room, whence it will gradually sink through the lighter, warmer air; and this, with fire-ventilation, will keep an ordinary room fresh and sweet, at least in winter-time.

Another method is to have the window open at the bottom, and to place, a couple of inches away from the opening, a screen somewhat higher than the bottom of the lower sash. A third way is to open the window from the top, and across the opening nail a piece of muslin or perforated zinc. Both these methods give an additional current of air; but the screen in the one case and the perforation in the other prevent such a rush as to cause the patient to complain of cold. If he objects to one plan, try another; but never be satisfied with anything short of complete ventilation, at the same time being very careful to avoid all draughts.

To keep a sick-room at the proper temperature is another serious matter. From sixty to sixty-five is the ordinary temperature; but various diseases require modifications, and it is always well to ask the doctor what he wishes in this respect. The nurse should never trust to her own sensation, but get a thermometer, and hang it up near the patient's bed. The temperature of a room will often vary by several degrees in different parts, and the nurse's concern is that her patient shall be breathing the right degree of warmth, so the thermometer should be hung as near the bed as possible. Special care is needed at night, as the outside air will be considerably colder than by day, and the nurse will have to keep the fire proportionately larger. This and

keeping the fire clear demand no little attention, especially when the patient does not sleep well and wakes at the slightest sound. When this is the case, it is well to start the night with a supply of coal done up in separate bits of paper. These may be dropped on one by one with hardly a sound. If the fire requires to be poked, use a piece of stick with a quick decided movement, which is better than worrying the patient by stealthy efforts to move first one piece of coal and then another. Here, a properly fixed gas radiating stove would be serviceable.

If sleep is a necessity for the patient, and he sleeps on till the fire gets very low, one of the forms of patent fire-lighters will cause less noise than the ordinary wood. Ashes should never be allowed to accumulate, and a wooden shovel for removing them is quite a comfort in a sick-room.

In very warm weather, of course the fire must be dispensed with; and there are days even in this country when to keep the temperature cool is no slight difficulty. The window should be open both at top and bottom, to give as much current as possible, and the register of the chimney must not be closed. Agitating the air with a large fan and sprinkling the window-sill with water are cooling; but best of all is a large block of ice placed in the middle of the room on a strainer, with a vessel below to receive the drippings.

VERMUDYN'S FATE.

A TALE OF HALLOWEEN.

IN TWO CHAPTERS.—CHAPTER II.

‘OF the troop of figures who came flocking into that strange and mysterious house, I observed that some of those behind held more lights, though the room was bright enough already, while the foremost carried dishes. But I had no eyes for the meat and drink they brought, or for anything but a girl in their midst; and it was just the same with Vermudyn, I saw in an instant; only, whilst I was full of horror and a dread I couldn't shake off or overcome, Vermudyn felt no fear, no surprise; only an intense delight flushed his face with joy, and his eyes glittered, as he came forward eagerly to meet the girl, who, it seemed to me, was pale as death, with eyes that glowed like flame.

‘I think I never saw so colourless a creature to live and move—if indeed she did live. And her hair—redder, and yet more golden than the chain Vermudyn held—was coiled round her beautiful head in the same snaky folds. She never looked at me for an instant, but went straight to Vermudyn, and putting both her hands in his, said some words in a strange language that sounded like spoken music. It was the sweetest voice I ever heard, and the softest. He answered in the same tongue, laughing and clasping the hands she put in his. From that minute, he fell under her spell, and had no eyes or ears for anything save that strange white woman.

‘She poured out wine for him, and he drank it with feverish haste, still looking at her and holding her hand. I noticed, though, that he shuddered when she first touched him, as if her hands were icy-cold; but he felt that no longer; he was just mad-like and stupid, as a bird is with a snake. He could only watch her with wild eyes that never left her face.

‘The men and women who came in with this awful, beautiful creature were dark mostly, and reminded me of the gipsies I used to see when I was a boy at home in England. One of the girls, dressed in outlandish clothes,

embroidered in scarlet and gold, came up and offered me some wine—even held it to my lips—and the scent of it made me mad to taste. The girl’s arm was close round my neck, and her wicked eyes, dry and bright as a toad’s, were looking into mine with a mocking smile, as she sang a soft, strange song, like laughing and crying all in one. But I shut my teeth hard, and turning away my head, closed my eyes, determined to resist her with all the strength of my will so long as I was conscious. Even in that dreamy, bewildered state, I felt afraid of entirely losing my senses, and something seemed to tell me I was lost if I yielded for an instant. My tempter laughed then, a loud hideous laugh, and flung down the silver cup she had offered me. The wine was spilt, and I fancied it turned to liquid flame as it touched the floor.

‘Still I had no power to speak or move from my place, but I watched Vermudyn more eagerly than ever. The supper-table was pushed on one side; and the room was now filled with dancers, dancing fiercely and madly to a wild tune, like the song of the gipsy when she leant over me with the poisoned wine. The tune rose louder and higher, and the dancers moved faster to keep time with the unearthly music—unearthly and wild, but so beautiful that I could have listened for ever, I thought. At times it sounded like the wind sighing through the aspens at night; then it rose to a roar like waves breaking on the beach in a storm. Yet, with all the changing sound, the roar of a storm, and the wailing of the wind—tears and laughter and pain—the music still kept time and tune, and the mad dance went on without a pause.

‘Foremost amongst them all was Vermudyn, and the woman in white with the glittering eyes and hair. He was holding her fast in his arms as they flew round; her head lay on his shoulder, and his face was bent down over hers. But I could see, as I watched him, that he had grown almost as white as the girl he held; and now her great eyes blazed with such awful light, I shuddered to look at them; while, as she danced and clasped Vermudyn, I fancied a tinge of colour came into her white lips, and her cheeks were a shade less deathly.

‘Still they danced, and still she grew brighter and warmer, but not like a living woman yet. And Vermudyn, like a mere straw drifting round and round in a whirlpool, became weaker and fainter every minute, and his face now was something ghastly to see; but his eyes were still fixed on the girl,

and he could see nothing and feel nothing beside. Her shining hair had got loosened in the dance, and seemed to be flying round them like thin golden flames as they moved.

‘It was she, now, who held Vermudyn up and forced him still to dance. His arms were round her yet; but her strength alone sustained the fainting man. She flew round as easily as ever; her feet scarcely touching the ground. The noise grew furious and deafening—music and laughter, shouts and screams that made my blood run cold, with snatches of old songs between, were all mingled together in one hideous mighty roar.

‘The faces of the men, or the demons who took their shape, got more fiendish as they danced; when suddenly the dancers swept out of the room in a wild crowd, just as they had entered it, and in their midst Vermudyn, lying dead, or senseless, on the floor. I tried to move—to reach him somehow at that desperate pass; but I couldn’t stir a finger. I struggled to shout aloud—to call his name. I might have been dead, for all the help I could give him. I had no power to speak or move.

‘Directly that demoniacal crew left the room, the lights seemed to fade and the fire grow dim. Thick darkness fell over everything, and I could not see a ray of light from where I now lay like a helpless log.

‘I remembered nothing more until I opened my eyes in broad daylight, stiff, and shivering with cold. I was lying at the entrance of a little cave among the rocks, wrapped in my blanket, and close to the embers of a dying fire. My horse, I saw, was picketed not far from me.

‘I was still in the Devil’s Panniken, sure enough. I saw the road by which we had come last night; but the place was strange to me; these were not the rocks I had seen before, which surrounded the place where we had spent the night.

‘We? I was quite alone now, and broad awake! The house and all else had vanished. As the recollections of the past night came crowding back, I sprang up and looked around in wonder. The house—the very room—in which I’d been was so distinctly before my mind’s eye, that I stood staring in amazement to find myself alone. No vestige of the house I’ve described to you, and no Vermudyn either! I told myself that I was clean mad. I

searched for him in a sort of frantic hurry, and shouted his name, but heard only the echoes answer me.

‘I tried to get farther into the cave at the mouth of which I’d been lying; but I soon found the way closed by a big chunk of rock. There was no other outlet to the cave, and there was nothing to explain the mystery. There was no sign of Vermudyn or his horse; *that*, no doubt, had strayed during the night. But where was *he*, and where, above all, had we two spent the night? I was fairly stunned. I felt for my knife, my revolver. These, with my belt, were safe enough. I had lost nothing. I was simply cold, hungry, and quite alone—save for my nag; and how glad I was of that companion, I can’t tell you! He would be the means of getting me away from that awful place faster than my legs could carry me.

‘I found a hunch of bread and some meat in my wallet; but I was too excited and wretched over Vermudyn’s disappearance, to light a fire and boil some tea. As soon as I’d swallowed down my breakfast, I mounted my horse, and rode backwards and forwards for a good two hours, searching for the body, for I was clear in my own mind that my poor old mate was dead.

‘Dead or alive, I hated to think of riding away and leaving him there in the Devil’s Panniken. But it was no good. I hunted every hole and corner within a mile of the place—as near as I could judge—where we had spent the night. At last I gave up the hopeless search—no signs of Vermudyn anywhere; and before noon, I had turned my horse’s head away from the wretched place, and for the first mile or so I rode so hard and fast that I began to blame my own folly in running away in broad daylight. From what, too?

‘Ay, there was the rub! What was I riding away from? and how had I escaped, while Vermudyn was lost? I was almost mad when I went over the past twenty-four hours. I couldn’t believe my senses. All I’d seen and heard too; and the only other witness was gone, vanished as completely as if he had been a spectre or part of some nightmare dream!

‘I felt my brain reel as I passed mile after mile along the lonely road, till at last I began to wonder if the Vermudyn I thought I knew was ever a living man, or if he made part of a long hideous dream, which I thought I should never forget or get over.

‘But I couldn’t cheat myself so; the man had written his name inside my pocket-book, “C. Vermudyn,” and had given me a ring he told me he once bought in an eastern bazaar. I’ve worn the ring ever since, in memory of him and that awful Halloween night.

‘Sure enough, Vermudyn was no dream; but from that day to this his name has never crossed my lips; and nothing would induce me ever again to ride through the Devil’s Panniken either by day or night.

‘In my own mind, boys, it’s as clear as daylight that the body found in that cave Gentleman Jack was telling you of a while since was neither more nor less than the skeleton of my poor old mate Vermudyn. I never thought to hear of his bones being found after all these years, poor old chap; or of telling you to-night what happened to us that Halloween in the Devil’s Panniken. I only hope he wasn’t alive in that awful place!—alive, and shouting for help, shut up there alone, and hopeless in the dark, whilst I was riding away in sunshine and clear air!—Phaw!’ muttered the old man; ‘it’s no good to think of that now; and talking’s dry work.—Another go of whisky, Pat!’

The murmurs of admiration, astonishment, and feeble doubt over this wondrous story of Old Grizzly’s were arrested almost ere they began, and each man stopped short, as a low, long laugh sounded through the room, and they then perceived what, being absorbed in the ‘tale of mystery,’ they had been too preoccupied to notice before—namely, that a stranger had entered the room some time during the progress of the narrative, and it was he who had dared to laugh! All eyes were turned significantly and inquiringly upon this presumptuous stranger; and one gentleman had gone so far as to deliver himself of the original remark, that ‘he calculated to call that mighty cool,’ when the new-comer advanced into the light of the flaring kerosene lamp, and Old Grizzly sprang to his feet, speechless and aghast.

‘Well, old boy, don’t you know me now?’ asked the stranger. ‘Am I so little like the Vermudyn you chummed with in Cherokee Dick’s claim?’

‘It’s Halloween *again*,’ muttered the other hoarsely, still delaying to take the proffered hand.

‘And an unlucky night for me to turn up, after the scurvy trick I played you,’ laughed the stranger. ‘But look here, mate—if you kept my ring, I’ve kept yours; and I’m flesh and blood safe enough—no spirit or demon, as you seem to fancy.’

Old Grizzly grasped both his hands, looking long and earnestly in his face meanwhile. ‘It is Vermudyn!’ he at last exclaimed. ‘Though how they found your bones yonder in the Devil’s Panniken, and yet you’re alive and hearty here to-night, is more than Pat Murphy or any other Irishman could explain!’

‘I had better say at once that there’s no mystery about this—this—gentleman’s arrival to-night, at least,’ interposed Gentleman Jack. ‘He is a chance companion and fellow-traveller of mine, and like myself, he hails from ’Frisco last.’

‘As you seem to be in the humour for telling stories to-night, mates,’ observed the newcomer, ‘perhaps it wouldn’t be amiss if I explained to my friend here, in your presence, the truth of his strange Halloween experiences on the night he parted company with me—or I with him—whichever you prefer.

‘I told you once,’ said he, addressing himself to Old Grizzly, ‘I had travelled a good deal and spent some years in the East; but I never told how much I had learned of the manner and customs of the people I lived with; or that, amongst other diverting knowledge, I acquired the art of smoking and eating that extract of hemp known in eastern countries as “hashish;” and no one save those who have been under its marvellous influence can ever understand the wonderful reality of the illusions it produces—stronger and more powerful than any opium in its effect, and less harmful to use. Years ago, the drug was almost unknown; to-day, there are “hashish” eaters and smokers in most of the big cities of the States.

‘At the time I’m speaking of, it was little known, and its effects scarcely understood. I had taken it often enough myself; but some idle whim prompted me to try the result of a dose on my friend here, that special and memorable night of which he has just told you something. Well, I administered a biggish dose in a pill I gave him for an aguish turn he’d had; and after that, as we rode along I let him have some tobacco, as his own was smoked out, and this tobacco of mine consisted almost entirely of the

dried hemp, the true “hashish.” We had not ridden a great way into the Devil’s Panniken, talking, as we rode, of the bad reputation of the place and the various legends concerning it, when the drug began to take effect on my old friend here, and he would have fallen from his horse, if I had not kept close beside him and supported him with my arm. As matters were then, I decided to dismount and camp for the night. For myself, I’d never been afraid of man or demon, and I knew my companion could go no farther; so I easily persuaded him to stop, though several times he muttered something about riding on.

‘Well, I wrapped him in his blanket like a babby, lighted him another pipe, just to compose him, and set to work to make a rousing fire, for the night was cold, and a keen frosty wind came sweeping down the ravine. He behaved strangely enough for some time, muttering and talking, while I watched by him; then by turns singing and laughing, while he stared at me or the fire. Once or twice he struggled hard to get up; but by-and-by the hashish overpowered him, and he slept soundly. I remained by him the whole night, and then tried in the early dawn to awaken him, as we wanted to push on. But he slept so heavily, that the idea occurred to me to ride off and leave him to wake alone, thoroughly mystified between his hashish visions and the loss of me!

‘It was a bad, mad sort of practical joke, but I was full of such follies in those early days. After I’d left him, I made tracks for the town we’d determined on visiting together, and waited for him some days; but he never turned up; and then an uneasy fear that some harm had befallen my friend through my own folly, got hold of me; and taking a sudden distaste for a digger’s life, I made my way to the nearest port, and went on board a ship just starting for Europe, and which, luckily for me, stood in need of an extra hand.

‘Since then, I’ve led a roving life on sea and shore, till fate landed me here to-night in time to listen to the account of my mysterious end, as it appeared to my worthy friend. I am sorry to spoil a good story, mates; but the pleasure two old chums experience in finding each other alive and hearty after so strange a parting—twenty years ago—will, I hope, in some degree compensate for your disappointment in discovering that the White Witch of the Devil’s Panniken had no hand in my fate after all!’

‘But,’ interrupted Gentleman Jack, ‘a skeleton with a ring on its finger was found recently in the cave.’

‘Quite possible,’ returned the new-comer; ‘but I am happy to say it is not that of Cornelius Vermudyn.’

QUEEN MARGARET COLLEGE.

CORRESPONDENCE CLASSES.

IN *Chambers's Journal* for October 25, 1879, we gave an account of a method of *Education by Post*, which has been the means of drawing considerable attention to the scheme. The scheme itself seems to be now in a flourishing condition, and bids fair to place the education of women on a sounder basis than heretofore. Some information regarding the progress and prospects of the Glasgow Association for the Higher Education of Women may not, therefore, be unacceptable.

The Association is now no longer known by its old cumbrous designation. It has risen to a higher level, is incorporated under the more euphonious name of Queen Margaret College, and looks forward to more extended operations than were possible in the first years of its existence. The munificent gift of a building in every way suited to the purpose to which it is to be devoted, has given a great impetus to the efforts to promote every branch of the work which was already undertaken. Queen Margaret College—the gift of a lady who from the first manifested a cordial interest in the higher education of her sex—stands within its own grounds, in a pretty, half-secluded spot not far from the University, and near enough to one of the great thoroughfares of Glasgow to be easily accessible to students from all parts of the city. Some progress has been already made towards the endowment of lectureships, and no doubt the liberality of the donor of the building will encourage the friends of education to make an effort worthily to complete what has been so generously begun. Meantime, lectures will be delivered by professors and others, tutorial classes will be held, and new schemes will be organised for the benefit of girls who have some respect for mental culture, and some aspirations towards the development of the faculties with which they have been endowed. As in the days of the ‘Association,’ so now the Correspondence Classes will take their place as a branch of the work of Queen Margaret College. There will be no change except in name.

A few years ago, comparatively little was known about Correspondence Classes, that is, of education conducted between teacher and taught through

the medium of the post-office. The system was on its trial. There were grave doubts and solemn shakings of the head when the scheme was suggested as a substitute for oral teaching. It was pronounced impossible that questions and answers sent to and fro between the teacher and the taught could produce any satisfactory result, though it was admitted by some objectors that this interchange might be of some use where other instruction was not to be had; it was better than nothing. Another class of objectors spoke deprecatingly of 'cram' with its train of evils, and among these were some who would have judged otherwise, had they only for a moment thought of what they were familiar with, university examination papers. One of the special advantages of Correspondence is that the pupils are obliged to study for themselves as thoroughly as they can any subject they take up. They receive a plan of the course so divided that they know exactly how much is expected for the lesson of each fortnight; they know where to look for information; books of study are prescribed; books of reference are suggested. Patient, careful, diligent study is the only true preparation for this kind of work, and the faculties of the pupil are fully exercised before the tutor steps in with corrections, comments, and criticism.

Preparation for university examinations was the primary object of the Correspondence Classes. To girls who had no opportunity of attending lectures or other classes, a way was opened by which they might compete for university certificates and prizes; and the high place taken by Correspondence pupils on the lists of successful candidates is sufficient proof of the efficiency of the system. But of incalculably greater, because wider, benefit are these classes to the ever-increasing number of young men and women who are not content with the small stock of knowledge acquired, under more or less favourable circumstances, at a period when the brain itself was still immature. There are many who thirst for knowledge, but know not how to direct their steps in the line of self-education. There is much misguided effort, leading only to disappointment and discouragement; sincere desire for improvement languishes, and finally passes away, just for want of guidance and stimulus. It is no wonder, then, that the system of Correspondence is rapidly growing in favour, and is carried on not only by Associations in connection with universities, but by private teachers, working either singly or in combination with others, under self-imposed regulations which are probably more elastic than those formed under the shadow of a university.

The scheme of Queen Margaret College combines the advantages of both, inasmuch as it offers instruction not only in the subjects prescribed for all the Glasgow University examinations which are open to women, but also in a number of subjects outside the University programme. In order to exhibit more clearly the nature and scope of the scheme, a brief review of the branches of study will be useful. They are classified in five grades. There are first, the preliminary or common subjects—English, history, geography, arithmetic, Scripture, and Latin. Next to these are what are termed the junior subjects—Composition, literature, history and geography, Scripture history, Latin, Greek, French, German, Italian, mathematics, astronomy, chemistry, botany, zoology, physiology, and physiography. The senior course includes, besides the subjects of the junior grade carried further, classes in political economy and logic. In the higher course the subjects are divided into five departments: (1) English, including the history of the language and literature; (2) foreign languages, with reference in each case to the history of the literature; (3) mathematical sciences; (4) logic, metaphysics, moral philosophy, political economy, and history; (5) chemistry, botany, geology, zoology, and physiology. The fifth course is intended to prepare candidates for the examinations in degree subjects. These subjects include all that are required for the M.A. and B.Sc. degrees.

In this large and comprehensive scheme there is provision made for a great variety of students, and it need scarcely be said that it attracts pupils at home and abroad, differing in age, capacity, and attainment. By means of the elementary classes, children are educated at home; and girls in the novitiate of their intelligence, who have come to the end of their school-days, find in them the means of culture. The literature of England, France, and Germany is open to them; studies in history and language, in science and philosophy, invite to further progress in what will enrich their minds, and save them from the vacuity that too often ensues when the routine of school-life is ended. Young men in business, ladies engaged in teaching, and ladies, too, with plenty of leisure for the pursuit of a favourite study, are among the most eager students; and not the least interesting are foreigners, whose papers call forth the hearty commendation of their tutors, not only for great painstaking and vigorous thinking, but also for a style of English which reflects great credit upon their powers of acquisition. These and many others find in the Correspondence Classes an aid and stimulus to study, and a medium of intercourse with men abreast of the age, taking a

fresh and living interest in the subjects which they teach, and sparing no pains to direct and encourage their students to honest, thorough, diligent, and therefore productive study.

It is scarcely possible to touch on a subject like this without endeavouring to enlist the active co-operation of the young people of the present day. Within the last few years many educational forces have been set in motion. By degrees the charge of flimsiness will be withdrawn from the education of girls; but it must always be kept in mind that anything worthy of the name of education is not to be got save at the cost of thorough systematic effort on the part of the student. Work begets the love of work, and what at first may be regarded as a drudgery, begins to be estimated at its true value, not only as a means to an end, but as in itself a pleasure. Subjects which educate thought and reflection are suggested to the pupil; the prospect widens; higher attainments are seen to be within reach; and an end is put to that easy contentment which is satisfied with a few showy accomplishments and a too slender knowledge of what is best worth knowing.

Detailed information relating to the Correspondence Classes may be had from the Honorary Secretary, Miss Jane S. Macarthur, 4 Buckingham Street, Hillhead, Glasgow.

THE MONTH: SCIENCE AND ARTS.

THE late terrible railway accident at Penistone—caused primarily by the breaking of the locomotive crank axle—has called attention to the fact that such breakage is by no means a rare occurrence, although it is seldom accompanied by fatal results. Some slight flaw in the metal, quite invisible on the outer surface, grows by constant vibration into a crack, and this crack eventually is the place of fracture. Although the accident has brought forth an unusual amount of comment by skilled engineers and others, we have seen no reference to a method of detecting flaws in metal which was discovered some years ago by Mr Saxton. He pointed out that a magnetic needle passed along such a bar would be deflected upon coming to a flaw. The method was experimented upon at the royal dockyards, and was found to give most certain results so far as bars of iron were concerned. Whether the system is applicable or not to railway axles, we do not know; but we call attention to the matter, as a possibly useful contribution to the subject under discussion. It is the opinion of many competent men that the above accident would not have been so disastrous if the train had been fitted with an automatic brake. It had what is called a continuous vacuum brake, which is effective enough so long as the coaches do not become separated. When such separation occurs, the wheels are no longer held in check. With the automatic brake, on the other hand, which is adopted by many of the leading railway Companies, the wheels are immediately acted upon, if by any means the coupling between the carriages should be broken. In the accident referred to, the train would with such a brake have been brought to a stand-still before it reached the point where it ran over the embankment.

An influential Committee has been formed with the endeavour to found a fund for the conservation of London antiquities. It seems that during recent building operations in the City, the discovery was made of some massive foundations evidently belonging to an important building of the Roman period. Several of the stones used were fragments of sculpture. These have now been preserved; but they ran a narrow escape of being again buried

where they were found. Similar discoveries in the metropolis are by no means rare, and the preservation of such relics should be provided for. The treasurer for the fund is Sir John Lubbock, M.P.

There is a certain region in the United States, reaching from the oil-wells of Pennsylvania to West Virginia, which has become known as the 'Gas Belt;' for wherever a well is sunk to a certain depth, the borer is rewarded for his pains by a liberal supply of natural gas, which can be utilised in heating, lighting, and other purposes. It seems that it is only of late years that the commercial importance of this phenomenon has been recognised. The Penn Fuel Company has been formed to bring the consumption of this gas into wider employment. There seems to be but two drawbacks to its use, one being unsteadiness of pressure, and the other a fear as to permanence of supply. The first difficulty might surely be obviated by mechanical means; and the second is hardly worth consideration, seeing that the yield of gas has been constant for many years, and as yet shows no sign of diminution.

A curious experiment dealing with another natural product has lately been made at Acqui by the proprietor of some baths there. This gentleman has at his disposal an inexhaustible supply of hot water from a natural spring, the temperature being a hundred and sixty-seven degrees Fahrenheit. The surplus not required for the baths has been diverted so as to flow through pipes to a garden on the outskirts of the town. Here the warm liquid flows beneath a number of forcing-frames containing melons, tomatoes, asparagus, and other garden produce. The result is that a supply of these delicacies is ready for market at a very early period of the year, and when, therefore, they fetch high prices. Surely this system could be extended with profitable results. Even in this country, far away from active volcanoes, we have hot springs where the experiment could be tried.

It is no new thing to get benefit from volcanic products; indeed, some of these products are of great commercial value. At Vulcano, one of the Lipari Islands on the north coast of Sicily, there is a small factory which was started some years ago by a Scotch firm, where a number of men are engaged in collecting materials deposited continually round the various vents. These products consist chiefly of sulphur, ammonia, and boracic acid.

The introduction of real Chinese birds'-nest soup to Londoners, to which we adverted last month, may raise the question as to what material such nests can be made of. An English naturalist living at Yokohama has lately published a very interesting account of a visit which he paid to Gormanton Caves, which are situated amid the tropical forests of North Borneo. From these caves come the bulk of the nests of which the soup is made, and they are the only place in the world where they can be obtained in any quantity. The caves are of immense extent, and are several hundred feet in height. They are covered with nests, which are built by swallows and bats; the material being a soft fungoid growth, which incrusts the limestone in which the caves are formed. The yearly value of the nests taken is between five and six thousand pounds on the spot. The value when they reach China is of course very much more. It is perhaps as well, considering the expensive nature of the luxury and its scarcity, that the consumption is not likely to increase from its introduction into Britain. To our barbarian palates it is decidedly insipid.

For three centuries, Britain has been able to boast that her adventurous sons have penetrated farther towards the frozen north than the sailors of any other nation. She must now yield the palm to America. The interesting story of the rescue of the six survivors of the Greely Expedition—who at the moment of their discovery were listening to prayers for the dying read by one of their number—is only second in interest to the story of Sir John Franklin, whose fate was for so long hidden in mystery. It seems to be a general feeling that no more expeditions to the frozen regions should be attempted. The barren honour of having arrived at a place so inaccessible that nobody has been there before you, is hardly worth the risk of being slowly starved to death. The Greely Expedition originally numbered twenty-five persons, so that nineteen have perished. This is a heavy price to pay for geographical knowledge however valuable; but of the scientific value of the expedition few details are as yet published.

Lieutenant Brown of the United States' navy has compiled a long official Report for his government on the progress of the Panama Canal, which is not quite so hopeful as the subscribers would desire. He considers that a great portion of the work accomplished is theoretical rather than practical, and that what has been done has been too costly. He thinks it evident that the scheme cannot be accomplished within the estimated cost nor within the

stipulated time. Two leading problems are likely to baffle the engineers—one is, how to dispose of the sixty million cubic mètres of earth which must be cut from the hilly part of the isthmus; and the other is the difficulty of dealing with the river Chagres, which was to form part of the channel. In the dry season, this river is a sluggish stream; but after the rains, it is a foaming torrent carrying everything before it. There is also a probability of an epidemic of yellow fever, which is generally of a fatal type in the district.

In the course of two lectures lately delivered at the Health Exhibition by Dr Cobbold upon the subject of Parasites in Food, some very interesting facts came to light. With regard to parasites, he tells us that the dreaded trichinæ, about which so much alarm was created some years ago in connection with the consumption of foreign pork, cannot live after being subjected to a heat of one hundred and twenty-two degrees Fahrenheit, which temperature is of course far below that to which meat is subjected in ordinary cooking. Referring to the late mackerel scare, the lecturer said that the entozoa of this fish were perfectly innocuous to mankind whether they were swallowed alive or dead. There are altogether no fewer than fourteen different kinds of parasites which find their home in the mackerel. Speaking of vegetarianism, he said that it was a mistake to suppose that those who eschewed flesh-foods had any consequent immunity from diseases provoked by parasites; on the contrary, the most common parasite known in this country was a vegetable feeder which could easily be received into the system by carelessly washed salads, &c.

A Java correspondent of our contemporary, *Nature*, relates a curious instance of cannibalism among snakes which came under his notice. He had killed close to his house a snake of very deadly character. Upon examining it some time later he found, protruding from its mouth, the tail of another snake, which eventually turned out to be of the same species and only a few inches shorter than its host. The natives of the place gave it as their opinion that the two creatures had been fighting, and that the victor had swallowed the vanquished. Another correspondent of the same journal tells of a similar case which he saw in India.

It deserves to be placed on record that the University of London have for the first time conferred the high degree of D.Sc. upon a lady. Mrs Sophia Bryant, by whom this honour has been achieved, is the daughter of the Rev.

Dr Willock, late rector of Cleenish, Enniskillen, and Fellow of Trinity College, Dublin. Mrs Bryant has for some time held the position of mathematical mistress at the North London Collegiate School for Girls.

An interesting article upon a very curious subject is contributed by M. C. E. Brown-Sequard to the French journal *La Nature*. This article takes for its title 'Attitudes after Death,' and deals with the numerous instances, on the field of battle and in other situations, where dead bodies have been found—sitting on horseback in one instance, raising a cup to the lips in another, transfixed in the position last assumed when sudden death came upon them. One case is very remarkable. A brakesman on an American railway was shot by a guerrilla, who lay in ambush in a forest through which the train passed. As he was shot, the unfortunate man was in the act of putting on the brake. His body remained fixed, his arms and hands stiff on the brake-wheel, whilst the pipe he was smoking remained between his teeth. It was extremely difficult to make the corpse let go its hold. The writer of the paper points out that this fixture of the body is quite different from the ordinary rigidity of death; and he believes that it depends upon the production of a persistent muscular action, like the fixed spasm often seen in hysterical or paralytic subjects. It is an act of life, but the last one.

For a long time, and more particularly since telephones have come into common use, it has been seen that our telegraphic methods are open to very great improvement. At present, each letter of every word transmitted requires one or more distinct signals, either by right or left deflections of a needle, or, as in the Morse method, by dots and dashes. In Signor Michela's steno-telegraph, which bids fair to come into very extended use, this difficulty is obviated. It works on the phonetic system; that is to say, the various sounds which go to make up speech—be the language that common to any European country—are grouped into series and represented by certain signs, each word being, as it were, dissected into sound-values. The system is, in fact, that of a telegraphic shorthand. The transmitting instrument consists of two keyboards, each having ten keys, each key communicating with a style on the receiving instrument, which prints a sign representing a particular sound. With such an apparatus, a skilled operator can telegraph words as they fall from the lips of a speaker as readily as a shorthand reporter can write them down. The system has for some time been in use in the Italian Senate, and is now on an experimental trial in

Paris. Whether it prove to be the telegraph of the future or not, it most certainly is constructed on a correct basis. We propose shortly to notice it more fully.

An invention which is said to be largely used in America has lately formed the subject of some interesting and successful experiments in London. Introduced by Messrs G. H. Gardner & Co., Southwark Bridge Road, London, it is known as the Harden Hand Grenade Fire-extinguisher, and consists of a glass flask containing a chemical liquid, which, when the flask is broken, emits a copious supply of that enemy to combustion, carbonic acid gas. The experiments were of the usual type—miniature conflagrations being put out readily when a grenade was thrown upon them. The extreme simplicity of the system is one of its chief recommendations; for the flasks, ornamental in appearance, can be disposed throughout a house, and are then ready for immediate use, in case an incipient fire should break out. They therefore take the place of the cumbrous fire-bucket, which is too often, when wanted, found to be empty.

So much has been published relative to smoke abatement in our large towns, and so little has been actually accomplished towards the solution of the problem, that many are beginning to despair, and to believe that the evil must be allowed to continue. Factories, which are the chief offenders, have been to some extent dealt with by law, and are now supposed to consume their own smoke; but the private householder, who contributes no small share of the carbon sent into the atmosphere, has, even if he had the will, been almost powerless in the matter. A stove has just been invented which, it may be hoped, will put a different complexion on the subject. At the back of the grate is a receptacle for the coals, which, by the action of a loose vertical iron plate, are forced forward to be consumed, so that the fuel is partly coked before it reaches the front of the fire. By an ingenious arrangement, the products of combustion are not carried direct to the chimney, but are delivered beneath the grate. This perfect combustion stove is the invention of Mr H. Thompson, of 29 Marquess Road, Canonbury, London.

Most people will be glad to hear that the guardians of our national picture-galleries have at last consented to allow their art treasures to be copied by photography. Why this permission has been delayed so long is strange, for nearly every continental gallery has long ago distributed fac-similes of its

contents to willing purchasers. There is one advantage gained in the delay, for by modern processes every touch of the artist's brush may be faithfully portrayed in the copy, and, moreover, that copy is of a permanent nature. In front of the National Gallery, London, a temporary structure has been erected into which the pictures can be carried to be operated upon in a good light. By this means, a far more satisfactory result can be obtained than by carrying the camera to the pictures as they hang upon the walls.

We some months ago recorded the fact that a prize of five hundred pounds had been offered by Mr Ellis Lever for a new Safety-lamp, which must fulfil certain stringent conditions. The adjudicators—all well-known scientific men—have just reported upon the one hundred and eight lamps which were sent in for competition. Of these, four were electric lamps, no one of which approached fulfilment of the conditions of the award; the rest being oil-lamps. All those which fulfilled the preliminary requirements were experimented upon; and very few indeed remained when the more extreme tests were reached. But none of the lamps really embraced the whole of the conditions enumerated, so the adjudicators felt themselves unable to make the award to any. At the same time, they highly commend two which nearly fulfilled those conditions. One of them is called the Marsaut Lamp; and the other is the contrivance of Mr William Morgan of Pontypridd, which they say presents several good features of marked originality.

The success of the Royal Tapestry Works at Windsor, where so much excellent work is turned out every year, has stimulated others to endeavour to produce a material similar in appearance, without all the costly processes which makes the woven fabric so expensive. In London recently, an Exhibition has been opened of the works of English artists upon a material known as Gobelins tissue. The work is executed with the brush like an ordinary picture on canvas, but with an intention to imitate the work of the loom.

A rare phenomenon in these latitudes, a waterspout, was recently witnessed at Southwold. The wind at the time was changeable, and attention was directed to the strange manner in which certain dark clouds seemed to be driven first in one direction and then in another. At length these clouds united, and their mass formed a clearly defined edge some distance above the horizon. From this edge there suddenly shot down a narrow tongue of

cloud, which seemed to strike the sea above five miles from the shore. Swayed from side to side by the wind at first, it gradually grew into an enormous column of water, estimated to be nearly one hundred and fifty yards in diameter, the mass of foam at its base indicating the enormous velocity with which water was being poured from it into the sea. The waterspout remained for twenty minutes, when it disappeared as quickly as it came. It was fortunate that there were at the time no ships in the neighbourhood.

An exhibition of what is called 'sanitary and insanitary houses' has been opened at the Health Exhibition. The idea seems to be to arrange two houses, the one as it ought to be, and the other as it ought not to be, and thus to exhibit the two in strong contrast the one to the other, by which an opportunity will be given to visitors, and those who choose to take the trouble to exercise their wits, of gaining instruction upon a point which has never before been brought forward in this manner. The houses are so placed that visitors enter by the ground-floor of the insanitary house, and pass through its various rooms, where all its defects are carefully and plainly set forth; then, on reaching the top-floor, the visitor crosses over to the sanitary house and descends through it.

OCCASIONAL NOTES.

THE FRENCH CROWN JEWELS.

ACCORDING to a contemporary, we learn that the French crown jewels when valued just after the Revolution of 1789 were estimated at eight hundred and forty thousand pounds, and they consisted of seven thousand four hundred and eighty-two diamonds, five hundred and six pearls, two hundred and thirty rubies, one hundred and fifty emeralds, one hundred and thirty-four sapphires, seventy-one topazes, eight garnets, and three amethysts. They were stolen from the Treasury, in which they had been deposited, and only a very small portion recovered; but the purchases made by Napoleon and the Bourbon kings brought the total of the crown jewels up to nine hundred thousand pounds when they were valued in 1832. When a fresh inventory was taken in 1875, it was found that the crown jewels consisted of seventy-seven thousand four hundred and eighty-six stones, weighing over nineteen thousand carats, and a part of these will shortly be sold. It is a mistake, however, to suppose that at the impending sale all the objects of historical interest will be reserved, for many of the jewels which belonged to the Duchess Anne of Brittany, and became an appanage of the French Crown when she married Charles VIII., are to be disposed of, as also several articles bequeathed by Cardinal Richelieu.

TREATMENT OF DIARRHŒA AND CHOLERA.

The following instructions, issued to local authorities in Scotland by the Board of Supervision, and certified by Dr Littlejohn, Medical Officer of the city of Edinburgh, may be useful in the event of cholera occurring in this country:

Local authorities, where there are either no medical men, or only a few scattered over the country, should provide themselves with a supply of suitable remedies. Among these may be mentioned—(1) elixir of vitriol; (2) the lead and opium pill; (3) the aromatic powder of chalk and opium; (4) ordinary mustard.

It is, however, not only of importance that an attack of cholera should be properly treated before medical assistance is procured, but also that the diarrhœa which may be present for days before the serious symptoms present themselves, should be checked at once. This may generally be effectually accomplished by causing persons so affected, and who are usually very thirsty, to drink freely of cold water to which elixir of vitriol has been added in the proportion of half a teaspoonful of elixir to the tumbler of water. Should the diarrhœa, in spite of the above treatment, continue for, say, two hours, a lead and opium pill should be given, and the dose should be repeated every time after the patient has been affected by the diarrhœa. If the patient, from weakness, be unable to follow his usual employment, he should be put to bed—care being taken that the limbs are kept warm, and that the bed is kept dry by means of a sheet of oilcloth, gutta-percha, or mackintosh between the sheet and the mattress. Should the discharge present the appearance of rice-water, and should there be urgent vomiting, cramps of the limbs, together with general sinking or collapse, the case should be regarded as most serious; and in the absence of a medical man, mustard poultices should be applied to the stomach and chest for half an hour at a time, and should be followed either by fomentations with warm water, or by bran or porridge poultices on the same parts of the body. These mustard and soft poultices should be alternated from time to time. Meanwhile the limbs should be well rubbed with warm cloths, and the lead and opium pills regularly administered, as directed above.

This treatment may be advantageously employed for all persons above fifteen years of age. From ten to fifteen years, the only change recommended in the treatment is that half a lead and opium pill, instead of an entire pill, should be given as a dose. Below ten years of age, the aromatic powder of chalk and opium should be substituted for the pill, and may be administered in doses of one grain for each year of life. Thus, an infant of one year should have one grain for a dose; and under one year, half a grain; while a child of six years should have six grains. The treatment otherwise is the same—care, however, being taken in the case of children not to allow the mustard to remain *beyond ten minutes* in contact with the skin.

Should there be no hospital at the disposal of the local authority, and should the house of the patient consist of one or two apartments, the other

members of the household should be at once removed. The room in which the sick person is lying should as far as possible be cleared of furniture; and the other apartment, if any, should be devoted to the preparation of articles of food and to the residence of the attendants, limited in number to a day and a night nurse.

GRANTON MARINE STATION.—We have to acknowledge receipt of the following sum in behalf of the Granton Marine Station:

£ s. d.

Aug. 4. A Friend, 1 0 0

EN PASSANT.

A SIDELONG glance like April sunlight shining
Through drifting clouds, a moment rent apart—
A glance which reads with swift, occult divining
Fond thoughts deep hidden in the inmost heart.

A sudden flash of love-born radiance gleaming
From two dark melting orbs of liquid light,
Whose haunting beauty sets the fond soul dreaming
Of far-off, unattainable delight.

A passing word of greeting, sweetly spoken
By two sweet lips whose lightest word is dear;
A moment more, and lo! the spell is broken
While yet its charm is ling'ring on the ear.

* * * * *

Ten years ago, I watched a sunbeam falling
Athwart the shadows of a sombre way;
Now, 'mid the after-glooms its charm recalling,
I bless the spot whereon its brightness lay.

G. C. J.

The Conductor of CHAMBERS'S JOURNAL begs to direct the attention of CONTRIBUTORS to the following notice:

1st. All communications should be addressed to the 'Editor, 339 High Street, Edinburgh.'

2d. For its return in case of ineligibility, postage-stamps should accompany every manuscript.

3d. MANUSCRIPTS should bear the author's full *Christian* name, Surname, and Address, legibly written; and should be written on white (not blue) paper, and on one side of the leaf only.

4th. Offerings of Verse should invariably be accompanied by a stamped and directed envelope.

If the above rules are complied with, the Editor will do his best to insure the safe return of ineligible papers.

Printed and Published by W. & R. CHAMBERS, 47 Paternoster Row, LONDON,
and 339 High Street, EDINBURGH.

All Rights Reserved.

*** END OF THE PROJECT GUTENBERG EBOOK CHAMBERS'S
JOURNAL OF POPULAR LITERATURE, SCIENCE, AND ART, FIFTH
SERIES, NO. 35, VOL. I, AUGUST 30, 1884 ***

Updated editions will replace the previous one—the old editions will be renamed.

Creating the works from print editions not protected by U.S. copyright law means that no one owns a United States copyright in these works, so the Foundation (and you!) can copy and distribute it in the United States without permission and without paying copyright royalties. Special rules, set forth in the General Terms of Use part of this license, apply to copying and distributing Project Gutenberg™ electronic works to protect the PROJECT GUTENBERG™ concept and trademark. Project Gutenberg is a registered trademark, and may not be used if you charge for an eBook, except by following the terms of the trademark license, including paying royalties for use of the Project Gutenberg trademark. If you do not charge anything for copies of this eBook, complying with the trademark license is very easy. You may use this eBook for nearly any purpose such as creation of derivative works, reports, performances and research. Project Gutenberg eBooks may be modified and printed and given away—you may do practically ANYTHING in the United States with eBooks not protected by U.S. copyright law. Redistribution is subject to the trademark license, especially commercial redistribution.

START: FULL LICENSE

THE FULL PROJECT GUTENBERG™ LICENSE

PLEASE READ THIS BEFORE YOU DISTRIBUTE OR USE THIS WORK

To protect the Project Gutenberg™ mission of promoting the free distribution of electronic works, by using or distributing this work (or any other work associated in any way with the phrase “Project Gutenberg”), you agree to comply with all the terms of the Full Project Gutenberg License available with this file or online at www.gutenberg.org/license.

Section 1. General Terms of Use and Redistributing Project Gutenberg electronic works

1.A. By reading or using any part of this Project Gutenberg electronic work, you indicate that you have read, understand, agree to and accept all the terms of this license and intellectual property (trademark/copyright) agreement. If you do not agree to abide by all the terms of this agreement, you must cease using and return or destroy all copies of Project Gutenberg electronic works in your possession. If you paid a fee for obtaining a copy of or access to a Project Gutenberg electronic work and you do not agree to be bound by the terms of this agreement, you may obtain a refund from the person or entity to whom you paid the fee as set forth in paragraph 1.E.8.

1.B. “Project Gutenberg” is a registered trademark. It may only be used on or associated in any way with an electronic work by people who agree to be bound by the terms of this agreement. There are a few things that you can do with most Project Gutenberg electronic works even without complying with the full terms of this agreement. See paragraph 1.C below. There are a lot of things you can do with Project Gutenberg electronic works if you follow the terms of this agreement and help preserve free future access to Project Gutenberg electronic works. See paragraph 1.E below.

1.C. The Project Gutenberg Literary Archive Foundation (“the Foundation” or PGLAF), owns a compilation copyright in the

collection of Project Gutenberg electronic works. Nearly all the individual works in the collection are in the public domain in the United States. If an individual work is unprotected by copyright law in the United States and you are located in the United States, we do not claim a right to prevent you from copying, distributing, performing, displaying or creating derivative works based on the work as long as all references to Project Gutenberg are removed. Of course, we hope that you will support the Project Gutenberg mission of promoting free access to electronic works by freely sharing Project Gutenberg works in compliance with the terms of this agreement for keeping the Project Gutenberg name associated with the work. You can easily comply with the terms of this agreement by keeping this work in the same format with its attached full Project Gutenberg License when you share it without charge with others.

1.D. The copyright laws of the place where you are located also govern what you can do with this work. Copyright laws in most countries are in a constant state of change. If you are outside the United States, check the laws of your country in addition to the terms of this agreement before downloading, copying, displaying, performing, distributing or creating derivative works based on this work or any other Project Gutenberg work. The Foundation makes no representations concerning the copyright status of any work in any country other than the United States.

1.E. Unless you have removed all references to Project Gutenberg:

1.E.1. The following sentence, with active links to, or other immediate access to, the full Project Gutenberg License must appear prominently whenever any copy of a Project Gutenberg work (any work on which the phrase “Project Gutenberg” appears, or with which the phrase “Project Gutenberg” is associated) is accessed, displayed, performed, viewed, copied or distributed:

This eBook is for the use of anyone anywhere in the United States and most other parts of the world at no cost and with almost no restrictions whatsoever. You may copy it, give it away or re-use it under the terms of the Project Gutenberg™ License included with

this eBook or online at www.gutenberg.org. If you are not located in the United States, you will have to check the laws of the country where you are located before using this eBook.

1.E.2. If an individual Project Gutenberg electronic work is derived from texts not protected by U.S. copyright law (does not contain a notice indicating that it is posted with permission of the copyright holder), the work can be copied and distributed to anyone in the United States without paying any fees or charges. If you are redistributing or providing access to a work with the phrase “Project Gutenberg” associated with or appearing on the work, you must comply either with the requirements of paragraphs 1.E.1 through 1.E.7 or obtain permission for the use of the work and the Project Gutenberg trademark as set forth in paragraphs 1.E.8 or 1.E.9.

1.E.3. If an individual Project Gutenberg electronic work is posted with the permission of the copyright holder, your use and distribution must comply with both paragraphs 1.E.1 through 1.E.7 and any additional terms imposed by the copyright holder. Additional terms will be linked to the Project Gutenberg License for all works posted with the permission of the copyright holder found at the beginning of this work.

1.E.4. Do not unlink or detach or remove the full Project Gutenberg License terms from this work, or any files containing a part of this work or any other work associated with Project Gutenberg.

1.E.5. Do not copy, display, perform, distribute or redistribute this electronic work, or any part of this electronic work, without prominently displaying the sentence set forth in paragraph 1.E.1 with active links or immediate access to the full terms of the Project Gutenberg License.

1.E.6. You may convert to and distribute this work in any binary, compressed, marked up, nonproprietary or proprietary form, including any word processing or hypertext form. However, if you provide access to or distribute copies of a Project Gutenberg work in a format other than “Plain Vanilla ASCII” or other format used in the official

version posted on the official Project Gutenberg website (www.gutenberg.org), you must, at no additional cost, fee or expense to the user, provide a copy, a means of exporting a copy, or a means of obtaining a copy upon request, of the work in its original “Plain Vanilla ASCII” or other form. Any alternate format must include the full Project Gutenberg License as specified in paragraph 1.E.1.

1.E.7. Do not charge a fee for access to, viewing, displaying, performing, copying or distributing any Project Gutenberg works unless you comply with paragraph 1.E.8 or 1.E.9.

1.E.8. You may charge a reasonable fee for copies of or providing access to or distributing Project Gutenberg electronic works provided that:

- You pay a royalty fee of 20% of the gross profits you derive from the use of Project Gutenberg works calculated using the method you already use to calculate your applicable taxes. The fee is owed to the owner of the Project Gutenberg trademark, but he has agreed to donate royalties under this paragraph to the Project Gutenberg Literary Archive Foundation. Royalty payments must be paid within 60 days following each date on which you prepare (or are legally required to prepare) your periodic tax returns. Royalty payments should be clearly marked as such and sent to the Project Gutenberg Literary Archive Foundation at the address specified in Section 4, “Information about donations to the Project Gutenberg Literary Archive Foundation.”
- You provide a full refund of any money paid by a user who notifies you in writing (or by e-mail) within 30 days of receipt that s/he does not agree to the terms of the full Project Gutenberg™ License. You must require such a user to return or destroy all copies of the works possessed in a physical medium and discontinue all use of and all access to other copies of Project Gutenberg™ works.
- You provide, in accordance with paragraph 1.F.3, a full refund of any money paid for a work or a replacement copy, if a defect in the electronic work is discovered and reported to you within 90 days of receipt of the work.

- You comply with all other terms of this agreement for free distribution of Project Gutenberg™ works.

1.E.9. If you wish to charge a fee or distribute a Project Gutenberg™ electronic work or group of works on different terms than are set forth in this agreement, you must obtain permission in writing from the Project Gutenberg Literary Archive Foundation, the manager of the Project Gutenberg™ trademark. Contact the Foundation as set forth in Section 3 below.

1.F.

1.F.1. Project Gutenberg volunteers and employees expend considerable effort to identify, do copyright research on, transcribe and proofread works not protected by U.S. copyright law in creating the Project Gutenberg™ collection. Despite these efforts, Project Gutenberg™ electronic works, and the medium on which they may be stored, may contain “Defects,” such as, but not limited to, incomplete, inaccurate or corrupt data, transcription errors, a copyright or other intellectual property infringement, a defective or damaged disk or other medium, a computer virus, or computer codes that damage or cannot be read by your equipment.

1.F.2. LIMITED WARRANTY, DISCLAIMER OF DAMAGES - Except for the “Right of Replacement or Refund” described in paragraph 1.F.3, the Project Gutenberg Literary Archive Foundation, the owner of the Project Gutenberg™ trademark, and any other party distributing a Project Gutenberg™ electronic work under this agreement, disclaim all liability to you for damages, costs and expenses, including legal fees. YOU AGREE THAT YOU HAVE NO REMEDIES FOR NEGLIGENCE, STRICT LIABILITY, BREACH OF WARRANTY OR BREACH OF CONTRACT EXCEPT THOSE PROVIDED IN PARAGRAPH 1.F.3. YOU AGREE THAT THE FOUNDATION, THE TRADEMARK OWNER, AND ANY DISTRIBUTOR UNDER THIS AGREEMENT WILL NOT BE LIABLE TO YOU FOR ACTUAL, DIRECT, INDIRECT, CONSEQUENTIAL, PUNITIVE OR INCIDENTAL DAMAGES

EVEN IF YOU GIVE NOTICE OF THE POSSIBILITY OF SUCH DAMAGE.

1.F.3. LIMITED RIGHT OF REPLACEMENT OR REFUND - If you discover a defect in this electronic work within 90 days of receiving it, you can receive a refund of the money (if any) you paid for it by sending a written explanation to the person you received the work from. If you received the work on a physical medium, you must return the medium with your written explanation. The person or entity that provided you with the defective work may elect to provide a replacement copy in lieu of a refund. If you received the work electronically, the person or entity providing it to you may choose to give you a second opportunity to receive the work electronically in lieu of a refund. If the second copy is also defective, you may demand a refund in writing without further opportunities to fix the problem.

1.F.4. Except for the limited right of replacement or refund set forth in paragraph 1.F.3, this work is provided to you 'AS-IS', WITH NO OTHER WARRANTIES OF ANY KIND, EXPRESS OR IMPLIED, INCLUDING BUT NOT LIMITED TO WARRANTIES OF MERCHANTABILITY OR FITNESS FOR ANY PURPOSE.

1.F.5. Some states do not allow disclaimers of certain implied warranties or the exclusion or limitation of certain types of damages. If any disclaimer or limitation set forth in this agreement violates the law of the state applicable to this agreement, the agreement shall be interpreted to make the maximum disclaimer or limitation permitted by the applicable state law. The invalidity or unenforceability of any provision of this agreement shall not void the remaining provisions.

1.F.6. INDEMNITY - You agree to indemnify and hold the Foundation, the trademark owner, any agent or employee of the Foundation, anyone providing copies of Project Gutenberg™ electronic works in accordance with this agreement, and any volunteers associated with the production, promotion and distribution of Project Gutenberg™ electronic works, harmless from all liability, costs and expenses, including legal fees, that arise directly or indirectly from any of the following which you do or cause to occur: (a)

distribution of this or any Project Gutenberg work, (b) alteration, modification, or additions or deletions to any Project Gutenberg work, and (c) any Defect you cause.

Section 2. Information about the Mission of Project Gutenberg

Project Gutenberg is synonymous with the free distribution of electronic works in formats readable by the widest variety of computers including obsolete, old, middle-aged and new computers. It exists because of the efforts of hundreds of volunteers and donations from people in all walks of life.

Volunteers and financial support to provide volunteers with the assistance they need are critical to reaching Project Gutenberg's goals and ensuring that the Project Gutenberg collection will remain freely available for generations to come. In 2001, the Project Gutenberg Literary Archive Foundation was created to provide a secure and permanent future for Project Gutenberg and future generations. To learn more about the Project Gutenberg Literary Archive Foundation and how your efforts and donations can help, see Sections 3 and 4 and the Foundation information page at www.gutenberg.org.

Section 3. Information about the Project Gutenberg Literary Archive Foundation

The Project Gutenberg Literary Archive Foundation is a non-profit 501(c)(3) educational corporation organized under the laws of the state of Mississippi and granted tax exempt status by the Internal Revenue Service. The Foundation's EIN or federal tax identification number is 64-6221541. Contributions to the Project Gutenberg Literary Archive Foundation are tax deductible to the full extent permitted by U.S. federal laws and your state's laws.

The Foundation's business office is located at 41 Watchung Plaza #516, Montclair NJ 07042, USA, +1 (862) 621-9288. Email contact

links and up to date contact information can be found at the Foundation's website and official page at www.gutenberg.org/contact

Section 4. Information about Donations to the Project Gutenberg Literary Archive Foundation

Project Gutenberg™ depends upon and cannot survive without widespread public support and donations to carry out its mission of increasing the number of public domain and licensed works that can be freely distributed in machine-readable form accessible by the widest array of equipment including outdated equipment. Many small donations (\$1 to \$5,000) are particularly important to maintaining tax exempt status with the IRS.

The Foundation is committed to complying with the laws regulating charities and charitable donations in all 50 states of the United States. Compliance requirements are not uniform and it takes a considerable effort, much paperwork and many fees to meet and keep up with these requirements. We do not solicit donations in locations where we have not received written confirmation of compliance. To SEND DONATIONS or determine the status of compliance for any particular state visit www.gutenberg.org/donate.

While we cannot and do not solicit contributions from states where we have not met the solicitation requirements, we know of no prohibition against accepting unsolicited donations from donors in such states who approach us with offers to donate.

International donations are gratefully accepted, but we cannot make any statements concerning tax treatment of donations received from outside the United States. U.S. laws alone swamp our small staff.

Please check the Project Gutenberg web pages for current donation methods and addresses. Donations are accepted in a number of other ways including checks, online payments and credit card donations. To donate, please visit: www.gutenberg.org/donate.

Section 5. General Information About Project Gutenberg electronic works

Professor Michael S. Hart was the originator of the Project Gutenberg concept of a library of electronic works that could be freely shared with anyone. For forty years, he produced and distributed Project Gutenberg eBooks with only a loose network of volunteer support.

Project Gutenberg eBooks are often created from several printed editions, all of which are confirmed as not protected by copyright in the U.S. unless a copyright notice is included. Thus, we do not necessarily keep eBooks in compliance with any particular paper edition.

Most people start at our website which has the main PG search facility: www.gutenberg.org.

This website includes information about Project Gutenberg, including how to make donations to the Project Gutenberg Literary Archive Foundation, how to help produce our new eBooks, and how to subscribe to our email newsletter to hear about new eBooks.